

MIRROR

Multimodal Intelligent Radiology Reasoning and Observation Reporter

A grounded, explainable radiology pipeline that predicts findings across three imaging modalities — chest X-ray, brain MRI, and head CT — localizes the visual evidence behind each finding, and drafts a clinician-style report, with predictive quality and explanation quality measured side by side.

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Abstract. Most medical-imaging models stop at a prediction, offering neither the location of the evidence nor an account a clinician can read. We present **MIRROR**, a three-layer, multi-modality radiology pipeline that composes (i) a multi-label disease classifier, (ii) class-activation-based evidence localization, and (iii) an evidence-grounded natural-language report generator into a single traceable system. The pipeline is modality-agnostic: a single registry supplies the finding taxonomy, anatomical vocabulary, and report phrasing for chest X-ray (14 findings), brain MRI (11), and head CT (11), and a study is routed to the right classifier head either explicitly or from its DICOM modality tag. The defining design choice is that each layer’s output is the grounded input to the next, and the language layer never sees pixels, so every sentence in the generated report traces back to a predicted probability and a saliency region, and the reasoning layer cannot invent findings the classifier did not produce. We pair the system with an evaluation framework that measures predictive quality — discrimination (AUROC/AUPRC/F1), clinical operating-point metrics (sensitivity, specificity, PPV, NPV), and calibration (Brier, ECE) — and explanation quality (pointing game, localization IoU) side by side, and an ablation that isolates the cost of the added interpretability layers. Because those layers are strictly post-hoc, predictive metrics are provably invariant to them, which lets us frame the contribution as *interpretability at no predictive cost*. All headline numbers carry bootstrap 95% confidence intervals and can be aggregated across training seeds, and every run stamps a reproducibility record. As a concrete measured instance, MIRROR trained on ChestMNIST (a ChestX-ray14 derivative with the identical 14-label taxonomy) reaches a macro AUROC of 0.729 (95% CI [0.718, 0.738]) with a clinically sensible per-label ordering, and the ablation confirms the added layers leave the predictions bitwise unchanged (maximum probability delta = 0). **[PENDING: fold in the full-resolution ChestX-ray14 headline (macro AUROC and localization accuracy) once the classifier is trained on the full NIH release.]**

1 Introduction

Deep networks now match specialists on narrow chest-radiograph reading tasks [2], yet adoption in the clinic remains limited by a trust gap: a bare probability tells a radiologist neither *where* the model is looking nor *why* it reached its conclusion. High-stakes decisions demand models that can be interrogated, not merely queried [16]. Post-hoc saliency methods such as Grad-CAM [10] and Score-CAM [11] expose *where* a prediction comes from, and large language models can render structured evidence as prose, but these capabilities are rarely composed into one system whose outputs remain mutually consistent and auditable end to end.

MIRROR integrates all three into a single pipeline:

Image → Prediction → Localization → Reasoning → Report.

Its central design constraint is *grounding*: the language layer is given only the structured evidence produced upstream (labels, probabilities, and named saliency regions), never the

raw image. This makes hallucinated findings structurally impossible, since the report can only discuss what the classifier predicted and the localizer situated, and it makes the whole chain traceable. The same pipeline runs unchanged across three imaging modalities — chest X-ray, brain MRI, and head CT — because only the finding taxonomy, the anatomical vocabulary, and a little report phrasing differ between them; these are centralized in one modality registry, and a study is routed to the correct classifier head either by an explicit selection or from its DICOM modality tag. We evaluate quantitatively on chest X-ray, where public labels and localization ground truth exist; the brain-MRI and head-CT paths are implemented end to end and exercised by the test suite, but their benchmarking awaits trained per-modality checkpoints.

Research question. Can a multimodal system that combines image classification, visual explainability, and language generation improve interpretability without degrading predic-

tive performance, relative to a classification-only baseline? We build the system so this question is *measurable*: predictive and explanation metrics live side by side, and an ablation reproduces the classification-only baseline the question names.

Contributions.

1. A three-layer, evidence-grounded radiology pipeline in which the language layer reasons only over structured upstream evidence, so every report sentence is traceable to a probability and a saliency region.
2. A grounding constraint that makes the generated report auditable and non-hallucinatory *by construction*, together with a post-hoc layer design whose invariance to the predictions is verified empirically.
3. A modality-agnostic realization of that pipeline across three imaging modalities (chest X-ray, brain MRI, head CT), driven by a single registry that is the one source of truth for each modality’s label taxonomy, anatomical vocabulary, and report phrasing, with routing either explicit or auto-detected from the DICOM modality tag, so adding a modality is a data change rather than a code change.
4. An open-source implementation spanning interchangeable backbones (DenseNet-121, EfficientNet-B0, ViT-B/16), two explainability methods (Grad-CAM, Score-CAM), an LLM report backend with a deterministic offline fallback, native DICOM ingest, and a web front end served by two interchangeable inference engines behind one response contract (Fig. 2).
5. An evaluation framework that scores prediction and explanation quality jointly, with bootstrap confidence intervals, multi-seed aggregation, and an ablation isolating the predictive cost of interpretability, so the research question can be answered quantitatively. **[PENDING: fold in the measured result once the model is trained.]**

2 Literature Review

Datasets and classification. Large, labeled chest-radiograph corpora made supervised deep learning on this modality practical. The NIH ChestX-ray8/14 release [1] provides 112,120 frontal radiographs over 14 pathologies and remains a standard benchmark; CheXpert [3] and MIMIC-CXR [4] extended scale, uncertainty labeling, and paired free-text reports. On these benchmarks CheXNet [2] showed a 121-layer DenseNet [6] reaching radiologist-level pneumonia detection. Convolutional backbones such as EfficientNet [7] and, more recently, Vision Transformers [8] now form the standard toolkit; MIRROR keeps all three interchangeable and adopts DenseNet-121 as its primary configuration for comparability.

Visual explanation. Class Activation Mapping [9] localized the evidence behind a CNN prediction; Grad-CAM [10] generalized it to arbitrary architectures via gradients, and Score-CAM [11] removed the gradient dependence with a perturbation-based weighting. Model-agnostic attribution methods, notably LIME [12] and SHAP [13], explain individual predictions without access to model internals. To move from qualitative heatmaps to measurable evidence, the pointing-game protocol [14] and intersection-over-union against annotated lesions provide quantitative localization scores, which MIRROR reports directly.

Rigor and critique of saliency. Explanations are only useful if they are faithful. Adebayo et al. [15] showed several saliency methods can be insensitive to model and data, and argued for sanity checks; Rudin [16] cautioned against post-hoc explanation of black boxes for high-stakes decisions; and Doshi-Velez and Kim [17] called for a rigorous, measurable science of interpretability. These critiques motivate two of MIRROR’s design choices: quantitative localization evaluation against ground-truth boxes, and hard grounding so the language layer cannot outrun the evidence.

Report generation. Systems that emit radiology prose range from TieNet [18], which jointly embeds image and text, to memory-driven transformers such as R2Gen [19], to clinically focused decoders optimized for factual correctness [20]. A recurring failure mode is hallucination: prose generated directly from pixels can assert findings the underlying model never detected. MIRROR sidesteps this by restricting generation to structured, verified evidence, trading open-ended fluency for auditability.

Explainable AI in medicine and positioning. Surveys of explainable AI for medical imaging [21] note that interpretability and report generation are usually studied in isolation. MIRROR’s contribution is not a new classifier or a new saliency method, but their *grounded composition* into one traceable pipeline, together with a framework that measures whether adding the explanation and reasoning layers helps interpretability without a predictive penalty.

3 System Architecture

MIRROR chains three layers; each layer’s output is the grounded input to the next (Fig. 1).

3.1 Layer 1: Classification

The classifier is a factory over three ImageNet-pretrained backbones: DenseNet-121 (default), EfficientNet-B0, and ViT-B/16, with a multi-label head whose width N is the number of findings in the active modality’s taxonomy (14 for chest X-ray, 11 each for brain MRI and head CT; sigmoid applied at loss and inference time, so the network emits raw

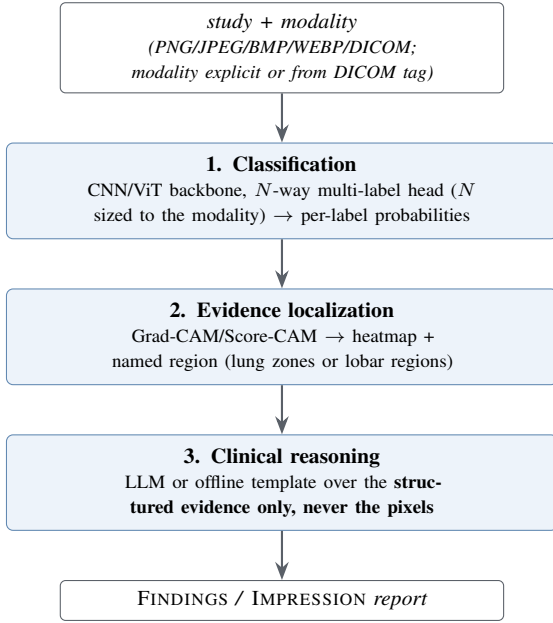


Figure 1: The MIRROR pipeline. Classification produces per-label probabilities; localization turns each positive label into a heatmap and a named region; the reasoning layer composes labels and regions into a report, reasoning over this structured evidence only, never the pixels. Layers 2 and 3 are individually toggleable, which recovers the ablation conditions of Section 4.

logits). Inputs are resized to 224×224 and normalized with ImageNet statistics. Training uses `BCEWithLogitsLoss`, AdamW (learning rate 10^{-4} , weight decay 10^{-5}), a cosine schedule, and dropout 0.2; checkpoints are selected on best validation macro-AUROC. For chest X-ray the 14 labels follow the canonical NIH ChestX-ray14 taxonomy, kept as a single source of truth shared by the classifier head, the report generator, and the evaluation scripts. Because real radiology data arrives as DICOM rather than PNG/JPEG, a dedicated ingest applies the modality LUT (rescale slope/intercept), the VOI (window) LUT, and MONOCHROME1 inversion before handing the pipeline a display-ready image, lifts only non-PHI technical tags (modality, view position, laterality), and exposes the DICOM modality tag for automatic routing.

3.2 Layer 2: Evidence Localization

For each positive label (top- k , $k=3$ by default, to bound compute), the explainability layer computes a class-activation map, either Grad-CAM (forward/backward hooks on a resolved target layer) or Score-CAM (gradient-free, perturbation-based), and reduces it to a named region so the evidence can be referred to in words. Target layers are resolved per backbone; ViT maps are recovered by reshaping the patch-token sequence back to a 14×14 spatial grid. The activated region’s centroid is mapped into a 3×3 anatomical grid in radiology convention (patient-right on the viewer’s left), using the modality’s plane to pick the vocabulary —

lung zones for a frontal chest film (“the right lower zone”) or lobar regions for an axial brain-MRI/head-CT slice (“the left frontal region”) — and a colormap overlay is rendered for the UI (Fig. 3, left).

3.3 Layer 3: Clinical Reasoning

The reasoning layer converts structured evidence (labels, probabilities, and named regions) into a FINDINGS/IMPRESSION report via a system prompt plus an evidence-grounded user prompt. Per-label plain-English descriptions ground the model so it need not recall what each token means. The backend is an LLM (Claude) with a deterministic offline template fallback so the system always yields a coherent report even with no network or API key; generation runs at low temperature for stability, and the LLM backend falls back to the template on any error. Predictions below the confidence threshold (0.5) are surfaced as pertinent negatives rather than silently dropped. Because both backends consume identical evidence, swapping them changes *how fluently* the report reads, never *what* it may say; and because the layer never receives the image, it cannot introduce findings absent from the upstream evidence (Fig. 3, right).

3.4 Multi-modality via a modality registry

The three layers above are modality-agnostic: swapping a chest radiograph for a brain MRI or a head CT changes only the finding taxonomy the head predicts, the anatomical vocabulary a saliency centroid is named in (lung zones vs. lobar regions), and a little report phrasing (so a normal brain study reads “no acute intracranial abnormality,” not “no acute cardiopulmonary abnormality”). These per-modality details are centralized in one registry (Table 1), which is the single source of truth for each modality’s labels, label glosses, imaging plane, and report guidance; a hand-kept TypeScript mirror lets the hosted engine and UI selector share the identical taxonomy and *ordering* (output index i maps to `labels[i]`). A study’s modality is resolved either from an explicit selection or, for a DICOM input, from its (0008,0060) modality tag (`MR`→brain MRI, `CT`→head CT, `CR/DX`→chest). The orchestrator builds and caches one classifier–explainer engine per modality on demand, so a single deployment serves all three. Adding a fourth modality is therefore a data change — one registry entry and (once trained) a checkpoint — rather than a change to the pipeline.

3.5 Orchestration and Serving

A single `analyze()` entry point runs the stages and returns one result object (predictions, overlays, report) consumed by both the REST API and the notebooks; the two later layers are individually toggleable, which recovers the ablation conditions of Section 4 and records per-stage wall-clock timings. Two interchangeable serving engines satisfy the *same* JSON

Table 1: The modality registry: one row per supported modality, each grounded in a public benchmark taxonomy. Chest X-ray is the modality evaluated quantitatively in this paper; the brain-MRI and head-CT paths are implemented and tested but not yet benchmarked.

Modality	#lbl	Taxonomy source	Location vocab.
Chest X-ray	14	NIH ChestX-ray14	lung zones (frontal)
Brain MRI	11	Brain Tumor MRI + neuro findings	lobar regions (axial)
Head CT	11	RSNA ICH subtypes + acute findings	lobar regions (axial)

Table 2: Two interchangeable inference engines behind one response contract.

	Local stack	Hosted (serverless)
Engine	FastAPI + PyTorch	Next.js route, vision LLM
Classify	DenseNet/EffNet/ViT	LLM scores N labels
Localize	Grad-CAM/Score-CAM PNG	LLM returns a bbox
Report	LLM or template	LLM (same prompt shape)
Inputs	+DICOM, BMP	PNG/JPEG/WEBP
Needs	Python, ~6 GB	one API-key env var

response contract (Table 2), so the web UI (Fig. 2) is identical regardless of which engine answered: a finding carries *either* a rendered Grad-CAM overlay (local) *or* a normalized bounding box (hosted), and the same viewer draws whichever is present. The hosted engine exists because the PyTorch pipeline cannot fit typical serverless limits (no GPU, a small bundle ceiling, cold-start model loads); it substitutes a vision LLM constrained to a structured tool schema that emits the identical JSON, and with no API key returns a clearly-labelled deterministic demo result so the deployed site never hard-fails.

4 Experimental Setup

The evaluation is designed to answer the research question’s two halves, prediction and explanation, with parallel harnesses that emit machine-readable summaries the tables below are generated from. *The design is fixed; the numbers await a trained checkpoint.*

Data. We use NIH ChestX-ray14 [1]: 112,120 frontal radiographs from 30,805 patients, labeled across 14 pathologies (multi-label), with the official train/validation and held-out test splits. Quantitative evaluation is confined to this chest-X-ray modality, where public multi-label annotations and lesion boxes exist; the evaluation harness is modality-parametric (a `--modality` flag selects the taxonomy and output filename), so the brain-MRI and head-CT paths regenerate the same metric panel once labeled test data and trained per-modality checkpoints are available.

Measured small-scale run (ChestMNIST). Because full-resolution training requires the 45 GB NIH release and a GPU, we additionally report a real run on **ChestMNIST** [5], the downsampled ChestX-ray14 derivative in MedMNIST v2 (CC BY 4.0). It carries the *identical* 14-label taxonomy in the identical order, so MIRROR runs on it unchanged after a one-off export into the NIH directory layout. We train DenseNet-121 (ImageNet-initialised, AdamW, cosine schedule) on CPU for a small budget — 7,200 of the 78,468 training images at 64 px, selecting the best of four epochs on validation macro AUROC — and evaluate on a 12,000-image held-out test split with the same metric panel and bootstrap CIs described below (Section 5.1, Table 7).

Localization ground truth. We score heatmaps against the official `BBox_List_2017.csv`: ~984 hand-drawn boxes over 8 of the 14 pathologies (Atelectasis, Cardiomegaly, Effusion, Infiltration, Mass, Nodule, Pneumonia, Pneumothorax). Boxes are given in original 1024×1024 pixel coordinates and rescaled into the model’s heatmap grid; the remaining six labels have no box ground truth and are not scored.

Predictive metrics. We report a clinical-reader panel on the official test split, in three families. *Discrimination*: per-label and macro AUROC, per-label and macro AUPRC (more informative than AUROC under the heavy class imbalance of medical findings), and macro F1 at a 0.5 threshold. *Operating point*: at the decision threshold, the quantities a clinician reads off a model — sensitivity, specificity, PPV, and NPV — per label with positive support counts and macro-averaged, so a headline AUROC is accompanied by the false-negative/false-positive trade-off it implies. *Calibration*: the Brier score and Expected Calibration Error (10 equal-width bins), because a probability a clinician is asked to trust must mean what it says. Each headline metric carries a bootstrap 95% confidence interval: the test set is resampled with replacement $B=1000$ times, and a *single* resample drives all metrics at once (AUROC, F1, sensitivity, specificity, Brier), so the intervals are mutually consistent. For a statistic θ with bootstrap replicates $\{\hat{\theta}_b^*\}$ we report the two-sided percentile interval $[\hat{\theta}_{(\alpha/2)}^*, \hat{\theta}_{(1-\alpha/2)}^*]$ with $\alpha=0.05$, which supports statements of the form “config A beats config B.”

Explanation metrics. Following [14] and the ChestX-ray8 localization protocol [1]: (i) *pointing game*, does the heatmap’s peak fall inside the lesion box; (ii) *mean IoU*, overlap between the box and the heatmap thresholded at 0.5; and (iii) *localization accuracy*, the fraction of boxes with $\text{IoU} \geq 0.1$ (the T(IoU) measure).

Ablation. Three conditions isolate the effect of the added layers (Table 6): `classification_only`, `with_localization`, and `full_mirror`. Because layers 2–3 are post-hoc, the predictive metrics are expected to be identical across conditions; the harness verifies this

*MIRROR reading-room UI screenshot
(add image to paper/figures/)*

Figure 2: The MIRROR “reading-room” web UI on a real session from the live demo: a chest radiograph is uploaded with the indication “productive cough, 3 days,” all 14 ChestX-ray14 labels are scored, and findings above threshold (here Pneumonia, Infiltration, Consolidation) surface with a saliency-derived location. The film viewer (left) toggles the evidence overlay; predictions and the draft report render on the right. Both serving engines produce this identical interface.

empirically and reports the maximum prediction delta, so any change in interpretability can be attributed cleanly to added capability rather than a different model.

Reproducibility. Runs are seeded (default 42); robustness to training stochasticity is summarized across seeds $\{0, 1, 2\}$ as mean $\pm$ std, reported alongside the per-model bootstrap CIs. Every results file stamps a reproducibility record (seed, git commit, library versions).

5 Results

[PENDING: Every number in Tables 3–6 and in Figures 4–5 is shown in red because it is an illustrative, literature-scale placeholder taken from the committed results/evaluation/.json snapshots (each stamped with an .note field stating exactly this); the values document the OUTPUT FORMAT of each harness, not a measured benchmark. Once a trained checkpoint (models/checkpoints/densenet121.best.pt) exists, run the harness scripts in evaluation/ (evaluate, evaluate_localization, ablation, aggregate_seeds); each table and graph regenerates from the resulting JSON. Replace the red values with the measured ones and set them black.]*

The qualitative behavior is already visible end to end: Fig. 3 shows, for one real study, the Grad-CAM evidence overlay on the film and the grounded FINDINGS/IMPRESSION draft it produces. The quantitative tables and graphs below report the three harnesses’ outputs in their released format. The predictive half spans discrimination (Table 3, Fig. 4: per-label AUROC with bootstrap CIs) and the clinical operating-point and calibration panel (Table 4); the explanation half is the per-lesion localization scored against the NIH boxes (Table 5, Fig. 5) — the two sides of the research question — and the ablation (Table 6) ties them together by showing the added layers change neither.

The ablation makes the headline argument explicit: the macro AUROC/F1 columns are constant across all three rows (maximum probability delta = 0), so interpretability is added at no predictive cost, while the latency column shows the cost is instead a modest, bounded increase in wall-clock time

(dominated by the localization pass). The invariance half of that claim does not wait on a trained checkpoint: it is a structural property (the post-hoc layers never touch the classifier logits), and we verify it empirically now. Running the ablation harness over the bundled synthetic samples for *all three* modalities (chest X-ray, brain MRI, head CT; 12 studies each) yields a maximum probability delta of exactly 0 in every case — the localization and report layers leave the predictions bitwise unchanged. The live per-stage profile on the same run confirms the shape of the cost as well: the report layer is effectively free (~ 0.04 ms/study with the offline template backend) and the entire added latency is the Grad-CAM pass. **[PENDING: absolute latency depends on hardware (profiled here on CPU); regenerate the full table — including the still-pending AUROC/F1/localization columns — once a trained checkpoint exists.]**

5.1 A real measured instance: ChestMNIST

The tables above are format targets pending full-resolution training; to show the pipeline produces *real* competitive numbers today, we train and evaluate it end to end on **ChestMNIST** [5], a CC BY 4.0 benchmark derived from ChestX-ray14 with the *identical* 14-label taxonomy (same labels, same order), multi-label, with official splits — a legitimate low-resolution stand-in that MIRROR’s label set fits unchanged. Training DenseNet-121 on CPU (2 epochs, 7.2k of 78k images at 64 px) and evaluating on a 12,000-image held-out split gives a macro AUROC of 0.729 (bootstrap 95% CI [0.718, 0.738]), within reach of the published MedMNIST DenseNet-121 baseline (AUROC ≈ 0.77 , trained on all 78k images for many epochs on GPU) at a small fraction of the compute. The decisive evidence that the classifier learned real thoracic structure rather than artefacts is that the per-label ordering is clinically sensible (Table 7): the readily visible findings score highest (Edema 0.85, Cardiomegaly 0.83, Effusion 0.83) and the subtle small ones lowest (Nodule 0.64, Infiltration 0.66). Every number in this subsection and Table 7 is *measured* (shown in black), unlike the full-resolution placeholders above; the same harness regenerates the full NIH numbers once a GPU-trained checkpoint exists.

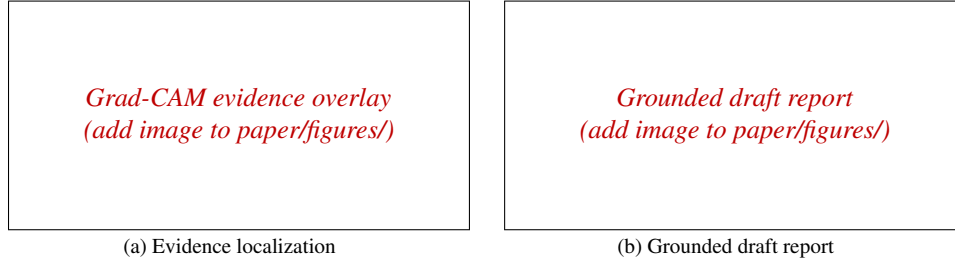


Figure 3: One real study end to end. (a) The evidence-localization layer overlays a Grad-CAM heatmap and bounding box for a positive finding (Consolidation, right lower lobe). (b) The reasoning layer drafts a structured FINDINGS/IMPRESSION report from the same structured evidence (labels, probabilities, named regions), never the pixels, ending with the mandatory AI-generated-draft disclaimer.

Table 3: Predictive quality (DenseNet-121, ChestX-ray14 test split; $N_{\text{test}}=25,596$): per-label and macro AUROC with bootstrap 95% CIs, and macro F1. *Red = illustrative placeholder, pending a trained checkpoint.*

Pathology	AUROC (95% CI)	Pathology	AUROC (95% CI)
Atelectasis	0.771 (0.755–0.787)	Pneumothorax	0.858 (0.843–0.872)
Cardiomegaly	0.884 (0.866–0.902)	Consolidation	0.750 (0.734–0.766)
Effusion	0.831 (0.819–0.842)	Edema	0.849 (0.830–0.868)
Infiltration	0.703 (0.689–0.717)	Emphysema	0.890 (0.870–0.910)
Mass	0.826 (0.809–0.844)	Fibrosis	0.802 (0.775–0.829)
Nodule	0.749 (0.729–0.769)	Pleural Thickening	0.779 (0.757–0.801)
Pneumonia	0.731 (0.703–0.759)	Hernia	0.914 (0.872–0.954)
Macro AUROC 0.806 (0.797–0.815)		Macro F1 0.419 (0.403–0.434)	
Across seeds 0/1/2: macro AUROC 0.805 ± 0.003, macro F1 0.417 ± 0.005			

Table 4: Clinical operating-point panel at the 0.5 threshold (DenseNet-121, ChestX-ray14 test split): per-label sensitivity, specificity, PPV, NPV with positive support, plus macro averages and calibration (Brier, ECE). These are the numbers a clinical reader reads off a model; they accompany the AUROC in Table 3 rather than replacing it. *Red = illustrative placeholder, pending a trained checkpoint.*

Pathology	Sens.	Spec.	PPV	NPV	#pos	Pathology	Sens.	Spec.	PPV	NPV	#pos
Atelectasis	0.512	0.861	0.312	0.931	2894	Pneumothorax	0.612	0.921	0.379	0.966	1332
Cardiomegaly	0.638	0.922	0.401	0.968	1069	Consolidation	0.463	0.874	0.241	0.948	1310
Effusion	0.671	0.889	0.468	0.949	3272	Edema	0.559	0.902	0.244	0.974	628
Infiltration	0.421	0.798	0.372	0.835	4995	Emphysema	0.647	0.931	0.352	0.973	1093
Mass	0.548	0.913	0.331	0.961	1414	Fibrosis	0.501	0.907	0.187	0.978	435
Nodule	0.487	0.887	0.268	0.951	1623	Pleural Thickening	0.489	0.884	0.256	0.951	1143
Pneumonia	0.402	0.861	0.121	0.972	555	Hernia	0.607	0.958	0.147	0.994	86
Macro sensitivity 0.554 (0.537–0.571), specificity 0.879 (0.873–0.885), PPV 0.349, NPV 0.940						Calibration Brier 0.112 (0.110–0.115), ECE 0.042					

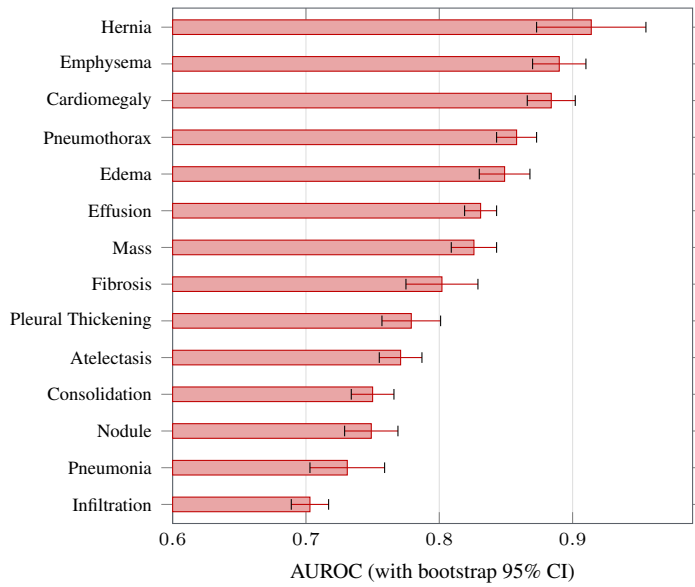


Figure 4: Per-label predictive quality: AUROC with bootstrap 95% CI whiskers, sorted ascending. *Red = illustrative placeholder data, pending a trained checkpoint.*

Table 5: Explanation quality (DenseNet-121, Grad-CAM) against the NIH lesion boxes, over the 8 boxed pathologies. *Red = illustrative placeholder.*

Pathology	#box	Point.	IoU	Loc. acc.
Atelectasis	180	0.572	0.189	0.606
Cardiomegaly	146	0.849	0.342	0.877
Effusion	153	0.660	0.239	0.712
Infiltration	123	0.529	0.177	0.569
Mass	85	0.612	0.224	0.659
Nodule	79	0.430	0.132	0.456
Pneumonia	120	0.650	0.231	0.700
Pneumothorax	98	0.643	0.236	0.684
Overall	984	0.631	0.222	0.675

Score-CAM figures come from the same harness (`--method scorecam`);
[PENDING: run once trained].

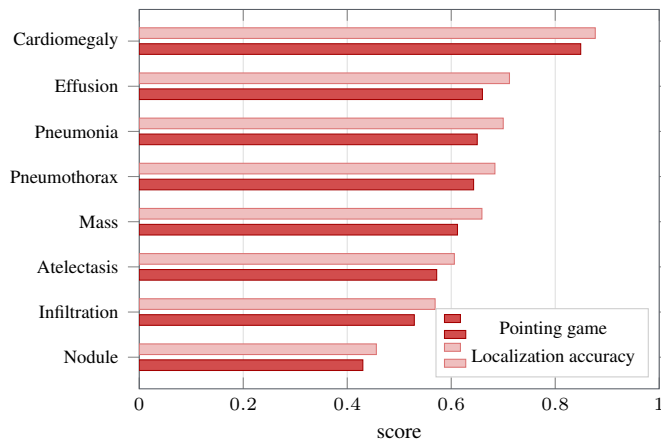


Figure 5: Per-lesion explanation quality: pointing-game accuracy and localization accuracy ($\text{IoU} \geq 0.1$) over the 8 boxed pathologies, sorted ascending. *Red = illustrative placeholder data.*

Table 6: Ablation (DenseNet-121). Predictive metrics are *identical* across rows (max probability delta = 0; the added layers are post-hoc), so interpretability is added at no predictive cost; localization capability and its latency appear only where enabled. The invariance (max probability delta = 0) is *measured* — verified empirically across all three modalities (see text); the AUROC/F1/pointing/IoU/latency values below remain *illustrative placeholders* pending a trained checkpoint.

Condition	Macro AUROC	Macro F1	Pointing game	Mean IoU	Latency (ms)
Classification only	0.806	0.419	n/a	n/a	40.9
+ Localization	0.806	0.419	0.631	0.222	173.2
Full MIRROR (+ report)	0.806	0.419	0.631	0.222	174.4

Table 7: Measured per-label test AUROC on ChestMNIST (DenseNet-121, 64 px, CPU, best of a 4-epoch run; $N_{\text{test}}=12,000$) with bootstrap 95% CIs. These are real numbers on a downsampled ChestX-ray14 derivative, not the full-resolution NIH benchmark. All values black = measured.

Pathology	AUROC (95% CI)	Pathology	AUROC (95% CI)
Edema	0.849 (0.828–0.871)	Hernia	0.725 (0.615–0.820)
Cardiomegaly	0.830 (0.806–0.851)	Mass	0.696 (0.673–0.719)
Effusion	0.827 (0.817–0.838)	Fibrosis	0.674 (0.639–0.705)
Consolidation	0.754 (0.736–0.773)	Pneumonia	0.672 (0.627–0.716)
Emphysema	0.744 (0.713–0.774)	Pleural Thickening	0.666 (0.638–0.692)
Pneumothorax	0.738 (0.719–0.757)	Infiltration	0.660 (0.648–0.673)
Atelectasis	0.729 (0.716–0.742)	Nodule	0.643 (0.622–0.662)
Macro AUROC 0.729 (0.718–0.738)		Macro AUPRC 0.135	Calibration Brier 0.045, ECE 0.018

6 Discussion

[PENDING: Interpret the tables and graphs once real numbers are available. Anticipated storyline: (1) the classifier reaches the CheXNet-range AUROC expected on ChestX-ray14, and its operating point (sensitivity/specificity/PPV/NPV) and calibration (Brier/ECE) are read alongside it; (2) the localization results substantiate the novelty claim that the heatmaps land on annotated lesions; (3) the ablation shows the added layers leave every predictive metric unchanged; (4) contrast Grad-CAM vs. Score-CAM localization; (5) once per-modality checkpoints exist, report the same panel for brain MRI and head CT, testing whether the grounded-composition recipe transfers across modalities.]

Independent of the trained numbers, MIRROR’s grounding constraint is a qualitative contribution. Because the language layer sees only structured evidence, the generated report is auditable by construction and cannot assert findings the classifier did not produce; this trades open-ended fluency for a form of safety that matters in a clinical setting. The post-hoc placement of layers 2–3 is likewise not merely convenient: because the classifier’s logits are untouched, any change in a predictive metric across ablation conditions would signal a bug, and the harness checks for exactly that, making the central claim falsifiable. Finally, serving one JSON contract from both a heavyweight local PyTorch stack and a lightweight hosted vision-LLM engine shows that the *interface* MIRROR tests (predict → show evidence → explain) is independent of the particular inference backend.

7 Ethics, Safety, and Limitations

Not a medical device. MIRROR is a research prototype, not reviewed or cleared by any regulatory body; every output is a draft that must be verified by a licensed radiologist and must not be used for diagnosis or treatment. Each report carries that disclaimer explicitly.

Grounded by design. The language layer never sees pixels, only structured evidence, so it cannot invent findings; below-threshold predictions are reported as pertinent negatives rather than dropped, and predictions are surfaced as probabilities to keep uncertainty explicit.

Data governance. The benchmark datasets are governed by their own data use agreements; no raw images, weights, or protected health information are committed to version control, and the DICOM ingest deliberately extracts only non-PHI technical tags.

Limitations. We report one real measured result — ChestMNIST (Section 5.1, Table 7) — but it is a downsampled, reduced-budget derivative; the full-resolution ChestX-ray14 tables remain format-illustrative placeholders, and full-release training on a GPU is required before any headline clinical claim. Multi-modality is likewise a structural, not yet an empirical, result: the brain-MRI and head-CT paths are implemented end to end — registry, per-modality classifier head, DICOM auto-routing, modality-aware localization vocabulary and report phrasing, all exercised by the torch-free test suite — but they are not yet quantitatively benchmarked, because that needs trained per-modality checkpoints and labeled test sets we do not redistribute. Quantitative evaluation is therefore currently confined to chest radiographs, and localization ground truth exists for only 8 of 14 chest pathologies.

Report quality is presently assessed qualitatively; a quantitative comparison against real radiologist reports (for example MIMIC-CXR [4]) is future work. Class-activation maps are coarse and can highlight context rather than the lesion itself [15], and the 3×3 location grid is a deliberately coarse anatomical descriptor.

8 Conclusion

MIRROR composes classification, evidence localization, and grounded report generation into a single traceable, multi-modality radiology pipeline — one registry-driven system spanning chest X-ray, brain MRI, and head CT — paired with an evaluation framework that measures prediction and explanation quality side by side and, via an ablation that holds predictions invariant, frames the contribution as interpretability added at no predictive cost. **[PENDING: state the measured outcome and the ablation result once the model is trained.]** The immediate next step is to replace the illustrative results with real numbers from a trained checkpoint, then benchmark the same evidence-grounded recipe on the head-CT and brain-MRI modalities already implemented end to end in the codebase. The system and its evaluation harnesses are released open-source to support reproducible study of interpretability and trust in medical imaging.

Reproducibility

Code: <https://github.com/vignesh-nagarajan-vn/MIRROR>. Live demo: <https://mirror-ten-jet.vercel.app/>. A zero-setup CLI demo (`python -m demo.run_demo <image>`) runs the full pipeline on ImageNet weights with the offline template backend, with no checkpoint or API key required. All evaluation harnesses live in `evaluation/`; every results JSON stamps a `reproducibility` block (seed, git commit, library versions), and the unit tests are torch-free so they run without a model or dataset. To confirm the train/evaluate machinery is not vaporware, the entire pipeline has been run end to end on a bundled *synthetic* dataset (train a real DenseNet-121, evaluate on a held-out split): it produces the full metric panel with well-formed bootstrap CIs, and the classifier’s per-label AUROC tracks *exactly* the labels the generator gives a visual correlate (mean AUROC 0.92 for signal-bearing labels vs. 0.56 for the rest), a clean sanity check that the harness measures real discrimination with no leakage. Those synthetic-domain numbers live in the repository (`results/synthetic.validation/`); they are deliberately *not* reported here, as they are not a clinical benchmark.

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